Robert Klement

Current position: Research Fellow at the European Southern Observatory (until 2026 Nov 14)
Incoming position: Postdoctoral Researcher, Georgia State University (from 2026 Nov 15)
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Github: https://github.com/robert-klement

Current & Previous Positions

Georgia State University (GSU), USA
Postdoctoral Researcher
2026 Nov 15 – (incoming)

European Southern Observatory, Chile
Research Fellow
2022 Nov – 2026 Nov

- 4th year of Fellowship (2025 Nov – 2026 Nov): 100% independent research, stationed at the CHARA Array, Mount Wilson, CA, USA; affiliated with Nice Observatory, France for SPICA instrument support
- Years 1–3 (2022 Nov – 2024 Nov): 50% observatory duties at VLT Paranal, Chile & 50% independent research; initially MATISSE Instrument Fellow, then GRAVITY instrument Fellow

Center for High Angular Resolution Astronomy (CHARA), Mount Wilson, CA, USA
Postdoctoral Research Associate
2017 Sep – 2022 Aug

- Night support for telescope array operations; science observing support for external proposals
- Alignment of the optical lab; participation in engineering upgrades
- Independent research & outreach activities

Charles University, Prague, Czech Republic
Doctoral Candidate / Researcher (part-time)
2012 Oct – 2017 Jun

- Independent research & student support
- Help with organization of international conferences

Education

Ph.D. 2012 Oct – 2017 Jun
Astrophysics · Astronomical Institute, Charles University, Prague, Czech Republic & European Southern Observatory, Santiago, Chile

M.Sc. (Ing.) 2011 Sep – 2014 Jun
Physical Engineering and Nanotechnology · Brno University of Technology, Brno, Czech Republic

M.Sc. (Mgr.) 2010 Sep – 2012 Jun
Theoretical Physics and Astrophysics · Masaryk University, Brno, Czech Republic

B.Sc. (Bc.) 2007 Sep – 2010 Jun
Applied Physics – Astrophysics · Masaryk University, Brno, Czech Republic

Harvard edX Professional Certificate - *Learning Python for Data Science* (issued 2026 Jan)

Research Interests

My research interests include the role of binarity in the evolution of massive stars, modelling of the circumstellar disks of classical Be stars including the tidal and radiative influence of stripped companions, and surveying the binarity of hot stars using optical/infrared interferometry, spectroscopy, Gaia astrometry, and radio interferometry. On the more technical side, I am interested in state-of-the-art technology and observing modes for optical interferometry: cameras, design, and science cases for beam combiner upgrades (*NSF proposal to be submitted this fall*). As for recent developments: machine learning, AI capabilities & application to astronomical research.

Granted Telescope Time, Observatory Work & Experience

Very Large Telescope (VLT) + Interferometer (VLTI)

- 20+ successful proposals for GRAVITY, PIONIER, UVES, XSHOOTER
- 240 nights as science observer for GRAVITY, MATISSE, PIONIER
- 50 nights as Night Shift Coordinator for the entire VLT + VLTI
- Daytime work using active optics to extend the brightness limit for XSHOOTER (published report)
- Expertise in data reduction & analysis for GRAVITY, PIONIER, UVES, XSHOOTER

- Assigned and completed observatory projects as Instrument Fellow (MATISSE and GRAVITY): (1) Updated the official internal documentation for VLTI and GRAVITY operations; (2) Created a new calibrator queue for GRA4MAT (GRAVITY fringe-tracking for MATISSE) low-resolution mode to avoid saturation and over-resolving; (3) Characterized the limitations of the GRAVITY dual-field Wide mode through analysis of hundreds of calibrator observations

The CHARA Array (chara.gsu.edu)
- 20+ successful observing proposals for MIRC-X, MYSTIC, VEGA
- 200+ nights as science observer for MIRC-X, MYSTIC, PAVO, VEGA, SPICA; 5 nights as array operator
- Expertise in data reduction & analysis for MIRC-X, MYSTIC, VEGA, SPICA

Very Large Array (VLA)
- 2 successful observing proposals; expertise in data reduction & analysis

Atacama Pathfinder Experiment (APEX)
- 5 successful observing proposals; 5 nights as science observer
- Expertise in data reduction & analysis from LABOCA bolometer

Lowell Discovery Telescope (LDT)
- 5 half-nights observing with EXPRES spectrograph

Apache Point Observatory
- Fully certified observer for ARC echelle spectrograph; 3 half-nights of observations

Hubble Space Telescope (HST)
- Several successful observing proposals as co-I for the STIS UV spectrograph

Community duties

- Executive member of the IAU Working Group on Active B-type stars (WGABS), 2022 Nov – present
- CHARA Array internal TAC member for observing semesters 2018B, 2019B, 2020B, 2022B, 2026B, 2027A
- Member of the Visitors Committee at ESO Chile (Nov 2022 – Nov 2025) - regular evaluation of applications for short-term research visits
- Evaluation of the technical feasibility of accepted observing proposals for GRAVITY (Nov 2022 – Nov 2025)
- Member of the Student Internship committee at ESO (2024 call)

Teaching & Mentoring

- Lecturer at the Workshop on New Visions in Optical Interferometry – interferometric binary grid search tutorial (2026 Mar, Socorro, NM, USA)
- Lecturer for a series of three 1.5-hour lectures: Introduction to Optical Interferometry: Basic Concepts, Observations, and Fitting of Geometrical Models – part of the Selected Chapters on Astrophysics at Charles University, Prague (2024 Nov, remote)
- Lecturer at the GSU summer lecture series 2021 – I gave a tutorial on observing, reduction, and data analysis for CHARA/VEGA (2021 Jul)
- Outreach presentation for Mt. Wilson Observatory docents about the CHARA Array (2019 May)
- Co-supervision of Ph.D. student: Suman Bhattacharyya during 1-year ESO Chile Studentship (2024 Jun – 2025 Jun)
- Supervision of Ph.D. student: Jana Švrčková during 1-year ESO Chile Studentship (2024 Nov – 2025 Nov)

Invited Review Talks

- *"Never be alone! The companions to Be stars"* · LIAC42 – To Be or not to Be: Understanding the star, the disk, and the stripped companions, Liège, Belgium, 13–17 July 2026
- *"Multiplicity among classical Be stars"* · 3,2,1: Massive Triples, Binaries and Mergers 2023, Leuven, Belgium, 17–21 July 2023
- I received two more invitations but could not attend (EAS 2023 and Hot Stars 2024: Life with Circumstellar Matter)

List of Publications

NASA/ADS: Published refereed papers - 33 refereed papers · 821 total refereed citations · Refereed h-index: 16 · 10 first-author refereed papers · 360 first-author refereed citations